

Discrete Mathematics — Worksheet 4: Induction and Counting

FlashTeX Course Staff

January 1, 1970

Name: _____ Section: _____

Part A: Warm-up (arabic labels)

1. Compute $\sum_{k=1}^5 k^2$.

2. Solve for x : $2x + 7 = 19$.

3. Show that $n^2 - n$ is even for every integer n .

Part B: Nested Structure

Work through the following multi-part induction problem. Each top-level item has lettered sub-parts, and one sub-part has roman-numeral sub-sub-parts.

1. Let $P(n)$ be the statement $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.

- (a) Verify $P(1)$.

- (b) State the inductive hypothesis $P(k)$.

- (c) Prove $P(k) \implies P(k+1)$ by completing each sub-step below.

- i. Write down $P(k + 1)$ explicitly.
- ii. Substitute the inductive hypothesis for the first k terms.
- iii. Simplify the resulting expression to match the closed form.

2. Let $Q(n)$ be the statement that $3 \mid (4^n - 1)$.

(a) Verify $Q(0)$.

(b) Prove $Q(k) \implies Q(k + 1)$, showing your algebra with the display below:

$$4^{k+1} - 1 = 4 \cdot 4^k - 1 = 4(4^k - 1) + 3.$$

Explain why this proves the divisibility step.

Part C: Compact Lists

1. State the Pigeonhole Principle in your own words.
2. Give one real-world example where it applies.

The following short list should have no extra vertical spacing between items:

1. $2 + 2 = 4$
2. $3 \times 3 = 9$
3. $\sqrt{16} = 4$

Part D: Terminology

Bijection

A function that is both injective and surjective, establishing a one-to-one correspondence between two sets.

Pigeonhole Principle

If n items are placed into m containers with $n > m$, then at least one container holds more than one item.

Well-Ordering Principle

Every nonempty set of nonnegative integers has a least element.

Part E: Checklist

- ☐ I verified the base case explicitly.

- ☐ I stated the inductive hypothesis clearly.
- ☐ I showed the inductive step follows logically.
- ☐ I double-checked my algebra in the displayed equations.

Part F: Challenge Problem

Prove, by induction on $n \geq 1$, that the sum of the interior angles of a convex polygon with $n + 2$ sides is $180n$ degrees. Use the space below, including at least one displayed equation such as

$$S(n + 1) = S(n) + 180,$$

to justify the inductive step.
